1. Pipeline Stages

Synthetic sample document for testing this application. The content is fictional.

A typical pipeline has five stages in order: build, unit tests, integration tests, deploy to staging and smoke tests. A failure in any stage stops the pipeline so that broken code never reaches the next environment.

Unit tests must finish in under ten minutes so developers get fast feedback. Artifacts produced by the build stage are stored once and reused by every later stage.

2. Flaky Tests

A flaky test passes and fails on the same code without any change. The team policy in these notes allows at most two automatic retries before a test is marked as failed.

A test that fails intermittently on three consecutive days is moved to a quarantine suite and a ticket is raised to fix its root cause. Typical causes are shared test data, timing assumptions and unstable external services.

3. Release Strategies

Blue-green deployment keeps two identical production environments and switches traffic from the old one to the new one at once. Canary release sends only five percent of traffic to the new version first and increases it if error rates stay low.

If the error rate rises, the release is rolled back within ten minutes. Feature flags let a team deploy code without exposing the feature until it is switched on.